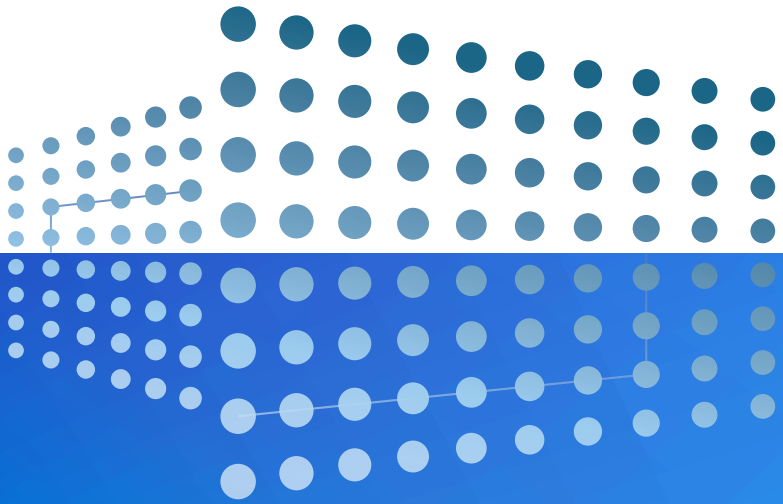
BlueNexus Technologies



Waterhaus™



OVERVIEW

Water Treatment Robot—Waterhaus™ redefines the traditional water plant construction model by applying an industrial product development mindset. Powered by AquaxGo™, the AI-driven intelligent operations platform independently developed by BlueNexus, Waterhaus™ features a highly integrated design that brings together all plant equipment and facilities as well as buildings and structures into a single, cohesive system. Manufacturing, commissioning, and trial operation are completed in the Waterhaus Super Factory, after which the water treatment robot is rapidly assembled and delivered on site—enabling fast deployment, standardized quality, and intelligent operation.

Leveraging insights and operational data from over 100 projects spanning 20 years, BlueNexus has transformed its deep expertise in water plant design, equipment manufacturing, engineering construction, and operations management into Waterhaus™—a fully productized intelligent water plant solution.

Waterhaus™ offers automated, safe and efficient management to deliver high-quality water for industrial enterprises, industrial parks and municipal applications. It effectively addresses challenges such as water scarcity, limited environmental carrying capacity, water safety risks, and high operating costs.



Waterhaus

APPLICATIONS

Waterhaus™ provides high-quality water for industrial enterprises, industrial parks and Municipality

Industrial Enterprises

Water Supply: Conventional water*/Non-conventional water**→ Treated by Waterhaus™→High-quality industrial water supply

Recycling: Regulation-compliant wastewater within/outside the factory→ Treated by Waterhaus™→High-quality industrial reclaimed water

Industrial Parks

Water Supply: Conventional water/Non-conventional water→ Treated by Waterhaus™→High-quality industrial water supply

Recycling: Regulation-compliant wastewater from industrial park→ Treated by Waterhaus™→High-quality industrial reclaimed water

Municipality

Water Supply: Conventional water/Non-conventional water→Treated by Waterhaus™→ High-quality drinking water/High-quality industrial water supply/Water source replenishment

Recycling: Regulation-compliant municipal wastewater→Treated by Waterhaus™→ High-quality reclaimed water/Water source replenishment

* **Conventional water:** Reservoir water, Groundwater, River water, Lake water
* **Non-conventional water:** Wastewater, Mine water, Brackish water, Seawater

ADVANTAGES

Modular Design

- 40-foot modular units with a highly integrated design, achieving up to 90% reduction in footprint area

Industrialized Production

- Industrialized production through factory assembly lines, ensuring strict quality control during commissioning, trial operation, and final acceptance

Rapid Installation and Delivery

- On-site assembly with rapid installation and immediate use, and cutting delivery time by 90%

AI-powered autonomous operation

- The water plant utilizes AI-driven multi-modal management, incorporating a domestic large language reasoning model for unmanned, safe, and efficient operation, reducing operational staff by 90% and reducing lifecycle costs by 50%

4S specialized service

- Developing a new operational and maintenance model combining AI agents with 4S on-site service, linking online intelligent operations with offline professional, real-time, and efficient maintenance services



PRODUCT SERIES

Product Name	Model	Standard Water Production m^3/d	Dimensions m(l*w*h)	Footprint m^2	Weight T
UR Series	Wh-UR-2500	2500	22.8x8.4x5.3	192	145
	Wh-UR-5000	5000	22.8×12.5×5.3	285	171
	Wh-UR-7500	7500	22.8×18.7×5.3	426	248
	Wh-UR-10000	10000	22.8×26.9×5.3	613	310
	Wh-UR-12500	12500	22.8×31.0×5.3	707	419

Model Description	Wh	UR	Water Source
	Waterhaus	Ultrafiltration & Reverse Osmosis process	Wastewater treatment plant effluent / Surface water

Product Name	Model	Standard Water Production m^3/d	Dimensions m(l*w*h)	Footprint m^2	Weight T
SW Series	Wh-SWS-3000	3000	22.8×12.5×5.3	285	110
	Wh-SWS-6000	6000	22.8×26.9×5.3	613	173
	Wh-SWD-2500	2500	22.8×14.6×5.3	333	251
	Wh-SWD-5000	5000	22.8×31.0×5.3	707	419

Model Description	Wh	SWS	SWD	Water Source
	Waterhaus	Seawater Desalination Single Stage	Seawater Desalination Dual Stage	Seawater

Product Name	Model	Standard Water Production m^3/d	Dimensions m(l*w*h)	Footprint m^2	Weight T
BW Series	Wh-BW-5000	5000	22.8x12.5x5.3	280	177
	Wh-BW-7500	7500	22.8x18.7x5.3	426	294
	Wh-BW-10000	10000	22.8x26.9x5.3	613	236
	Wh-BW-12500	12500	22.8x31.0x5.3	707	385

Model Description	Wh	BW	Water Source
	Waterhaus	Brackish Water	Brackish Water

Product Name	Model	Standard Water Production m^3/d	Dimensions m(l*w*h)	Footprint m^2	Weight T
UF Series	Wh-UF-3500	3500	22.8x6.4x5.3	146	84
	Wh-UF-7000	7000	22.8x8.4x5.3	192	121
	Wh-UF-10500	10500	22.8x10.5x5.3	239	158

Model Description	Wh	UF	Water Source
	Waterhaus	Ultrafiltration Process	Surface water

AquaxGo™

Overview

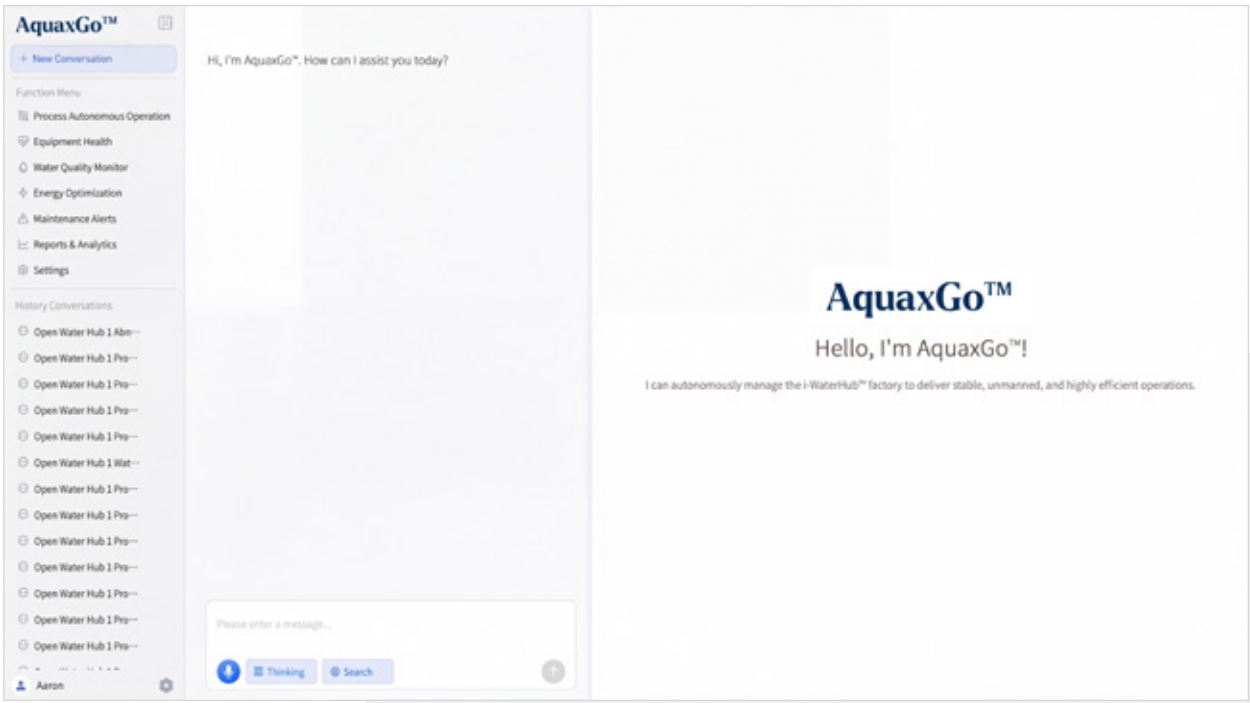
AquaxGo™ is an unmanned, AI-driven system that autonomously manages water plant operations, replacing routine human tasks. Operating within a closed-loop Online-to-Offline framework, it delivers safe, efficient, and cost-effective performance with minimal human intervention.

Features

- Acts autonomously—analyzes, decides, and executes tasks, not just a software tool
- BlueNexus self-developed process optimization models
- Computer vision, auditory sensing, and infrared for real-time environmental awareness
- Large language models enabling predictive maintenance of equipment and processes

Applications

Service center in Wuxi were efficiently managed by just 10 staff, leading to a 90% decrease in labor, 50% reduction in equipment failure rates, 15% lower chemical costs, 30% lower electricity consumption, and a 35% reduction in overall operational costs. This project also introduced an innovative distributed, unmanned operation management model for the region.



REFERENCES



Thermal Power

- **Wuxi Longting Waterhaus™**
  Capacity: 10,000 m³/d
  Users: Nengda Thermal Power Co., Ltd.
- **Waterhaus™ of Shijiazhuang Yuhua Thermal Power Plant, CHD**
  Capacity: 10,000 m³/d
  Users: Shijiazhuang Yuhua Thermoelectric Co., Ltd.



Electronics Manufacturing

- **Wuxi Jianding Waterhaus™**
  Capacity: 3,300 m³/d
  Users: Tripod (Wuxi) Electronic Co., Ltd.
- **Henan Lankao Waterhaus™**
  Capacity: 5,000 m³/d
  Users: Fulian Technology (Lankao) Co., Ltd.
- **Jiangsu Yancheng Waterhaus™**
  Capacity: 5,000 m³/d
  Users: Dongshan Precision Manufacturing Co., Ltd.



Clean Energy and Semiconductor

- **Wuxi Anzhen Waterhaus™**
  Capacity: 5,000 m³/d
  Users: Huasun New Energy Technology Co., Ltd.
  UtmoLight Technology Co., Ltd.
  XinDong Semiconductor Technology Co., Ltd.
- **Zhejiang Zhoushan Waterhaus™**
  Capacity: 5,000 m³/d
  Users: Zhejiang Winhitech New Energy Co., Ltd.

Advanced Material

- **Shanxi Yangqu Waterhaus™**
  Capacity: 2,000 m³/d
  Users: Shanxi Horizon New Material Technology Co., Ltd.





* Middle East Project Rendering

Drinking Water

- **Middle East Waterhaus™ Premium Drinking Water Project**
  Capacity: 20,000 m³/d
  Raw water source: Brackish groundwater

BlueNexus Technologies Pte. Ltd.

BlueNexus is a Singapore-based, technology-driven water solutions company focused on modular, scalable, and AI-powered systems. Founded in 2023, the company aims to transform conventional water infrastructure into intelligent, high-performance solutions—particularly in water reuse and decentralized treatment.

Its flagship product, Waterhaus™ and AquaxGo™, is prefabricated and digitally enabled water treatment system designed for both industrial and municipal applications. By combining advanced treatment processes with intelligent automation, BlueNexus is redefining how water systems are built and operated.

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